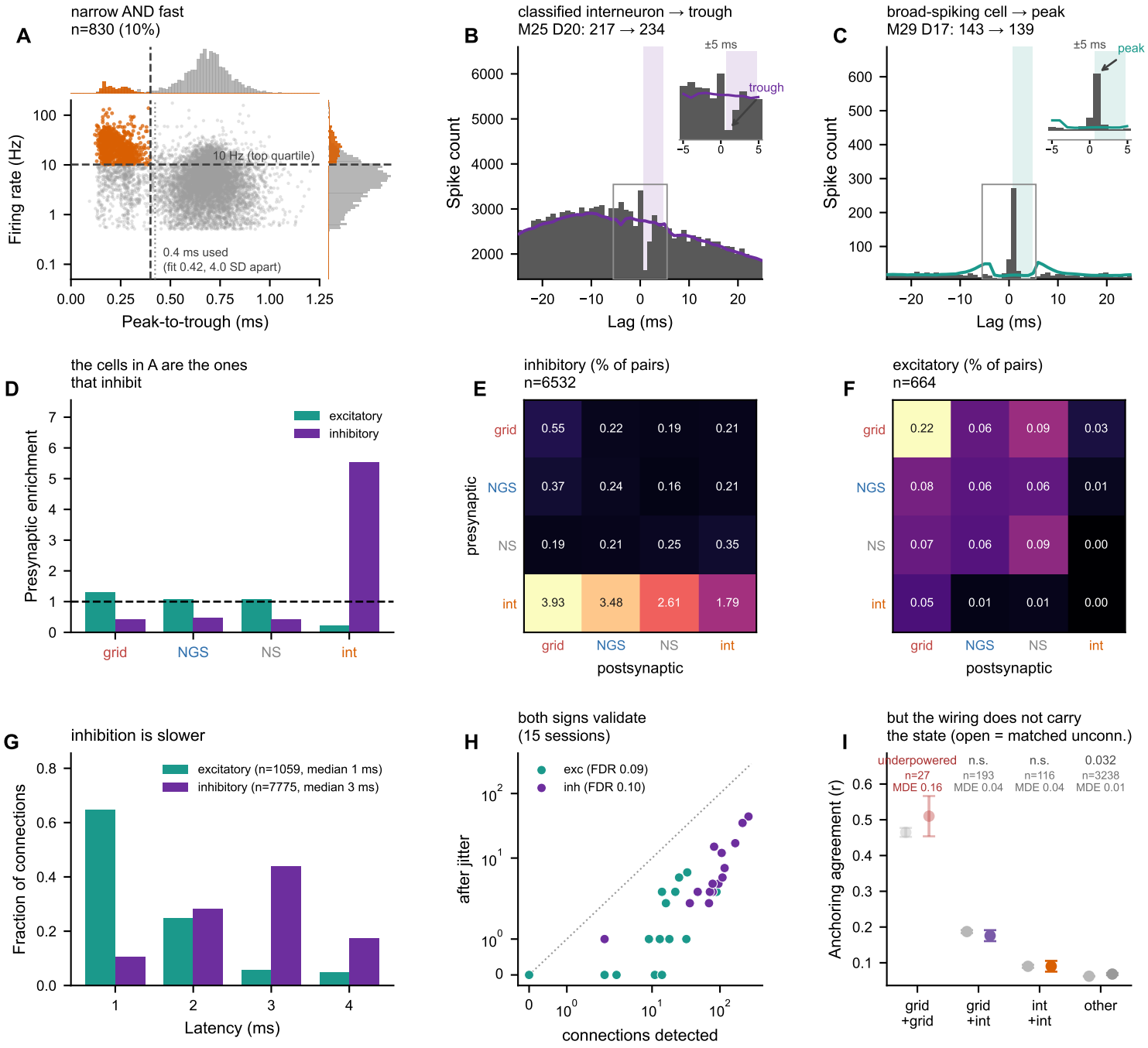

A
narrow AND fast
n=830 (10%)
Firing rate (Hz)
100
10
1
0.1
10 Hz (top quartile)
0.4 ms used
(fit 0.42, 4.0 SD apart)
0.00
0.25
0.50
0.75
1.00
1.25
Peak-to-trough (ms)
B
classified interneuron → trough
M25 D20: 217 → 234
Spike count
6000
5000
4000
3000
2000
±5 ms
trough
−5
0
5
−20
−10
0
10
20
Lag (ms)
C
broad-spiking cell → peak
M29 D17: 143 → 139
Spike count
600
500
400
300
200
100
±5 ms
peak
−5
0
5
−20
−10
0
10
20
Lag (ms)
D
the cells in A are the ones
that inhibit
Presynaptic enrichment
7
6
5
4
3
2
1
0
excitatory
inhibitory
grid
NGS
NS
int
E
inhibitory (% of pairs)
n=6532
presynaptic
grid
NGS
NS
int
0.55
0.22
0.19
0.21
0.37
0.24
0.16
0.21
0.19
0.21
0.25
0.35
3.93
3.48
2.61
1.79
grid
NGS
NS
int
postsynaptic
F
excitatory (% of pairs)
n=664
grid
NGS
NS
int
0.22
0.06
0.09
0.03
0.08
0.06
0.06
0.01
0.07
0.06
0.09
0.00
0.05
0.01
0.01
0.00
grid
NGS
NS
int
postsynaptic
G
inhibition is slower
Fraction of connections
0.8
0.6
0.4
0.2
0.0
excitatory (n=1059, median 1 ms)
inhibitory (n=7775, median 3 ms)
1
2
3
4
Latency (ms)
H
both signs validate
(15 sessions)
after jitter
exc (FDR 0.09)
inh (FDR 0.10)
10^2
10^1
10^0
0
0
10^0
10^1
10^2
connections detected
I
but the wiring does not carry
the state (open = matched unconn.)
Anchoring agreement (r)
0.6
0.5
0.4
0.3
0.2
0.1
underpowered
n=27
MDE 0.16
n.s.
n=193
MDE 0.04
n.s.
n=116
MDE 0.04
0.032
n=3238
MDE 0.01
grid
+grid
grid
+int
int
+int
other